

Methods of delivery to the pons

Diffuse intrinsic pontine glioma, H3K27-altered diffuse midline glioma, and related CNS diseases

Literature synthesis of 1,066 curated records (1989–2026)

Methods of delivery to the pons in DIPG and related diseases

Synthesis of **1,066** curated records (1989–2026) on physical delivery of therapeutics to the pons/brainstem. Per-modality detail is in [notes/](#); this report compares routes, grades their evidence, and states what is missing.

1. The problem statement

DIPG/DMG fails treatment for reasons that are at least partly logistical rather than pharmacological:

- The pontine blood-brain and blood-tumour barrier is largely preserved. Tumours are often non-enhancing, the neurovascular unit is comparatively intact (El-Khouly 2021, PMID 37284626), and there is evidence the tumour actively maintains barrier function (Angpt1 upregulation in H3K27M models, HGG-22 2021 abstract).
- ABCB1/ABCG2 efflux removes what does cross, so nominal brain concentrations overstate unbound exposure and delivery method and efflux status are not separable variables (Power 2023, PMID 37128506; Tsvankin 2020, PMID 31225627).
- The target has no mechanical tolerance. A few hundred microlitres of infusate deform the pons measurably (Bander 2020, PMID 31896089); oedema that would be trivial supratentorially is symptomatic here.
- Disease is diffuse and often extends beyond the pons or seeds the CSF, so a focal route cannot be the whole answer.

Consequence: trials that do not measure delivery cannot distinguish drug failure from delivery failure — the single most consistent theme across this literature.

2. Route-by-route assessment

Route	Best pontine evidence	Maturity	Main limitation
Convection-enhanced delivery (CED)	Phase 1/2 trials: ¹²⁴ I-omburtamab (Souweidane 2025, PMID 39969230), IL13-PE (Heiss 2019, PMID 30544335), MTX110 (Mueller 2023, PMID 37318058), carboplatin/valproate intermittent CED (Szychot 2021, PMID 33575829), ACNU phase 2 (Saito 2025, PMID 40150841)	Highest — repeated dosing via implanted catheters is achievable	Coverage of a diffuse target, reflux, short residence time, infusion-related toxicity (Hollingworth 2022, PMID 35933568)
Focused ultrasound BBB opening (FUS-BBBO)	Preclinical brainstem series (Alli 2018, PMID 29753957; Englander 2021, PMID 33753753; Ishida 2021, PMID 33188825; Martinez 2023, PMID 37795179), with RT (Tazhibi 2024, PMID 38555449); pediatric DMG trials (Wu 2025, PMID 41223245)	Rising fast — non-invasive and repeatable	Skull/acoustic access to the posterior fossa, transient opening window, does not defeat efflux, oedema budget
Intra-arterial ± osmotic BBB disruption	Historical pontine series (Fujiwara 1994, PMID 8108006; Hall 2006, PMID 16314949), balloon-assisted SIACI bevacizumab (Riina 2010, PMID	Feasible but stalled	Posterior-circulation catheterisation risk, non-selective barrier

Route	Best pontine evidence	Maturity	Main limitation
	20377982), SIACI with BBBD in children (McCrea 2021, PMID 34359048), MSC-virus delivery (Carceller 2014, PMID 24327128)		opening, modern data are mostly supratentorial
CSF routes (intrathecal / ICV / fourth ventricle / cisternal)	Fourth-ventricle infusion in piglet and NHP (Sandberg 2008, PMID 18447671; 2012, PMID 22546032); human brainstem CSF tracer distribution (Melin 2023, PMID 38063195); reservoirs/ports in practice	Mature as hardware, indirect for parenchyma	Penetration into pontine parenchyma is shallow; best for CSF-adjacent, disseminated disease and cell therapy
Locoregional cellular therapy	ICV B7-H3 CAR-T phase 1 in DIPG (Vitanza 2025, PMID 39775044) and non-pontine DMG (Ronsley 2026, PMID 42503899); IV→ICV GD2-CAR (Monje 2025, PMID 39537919; Jiang 2026, PMID 40682569)	Clinically active, route now standard-of-design	Local inflammatory neurotoxicity in the pons (Ronsley 2026, PMID 41798119), trafficking, anti-CAR immunity
Oncolytic viruses / vectors	Intratumoural DNX-2401 into the pons (Gállego Pérez-Larraya 2022, PMID 35767439); Ad-TD-nsIL12 (Qian 2025, PMID 40721414); MSC carriers (Chastkofsky 2021, PMID 33272983)	Proof of feasibility in patients	Single injection distribution, immune clearance, uncertain intratumoural spread
Nanocarriers	CED nanofiber depot (Bellat 2020, PMID 32301996); micellar panobinostat CED (Singleton 2017, PMID 28260886); exosomal panobinostat/PPM1D-siRNA (Shan 2022, PMID 35585670)	Preclinical	Reach patients only through another route; no pontine clinical trial of a nanocarrier per se
Intranasal	Nanoliposomal SN-38 (Sasaki 2023, PMID 36599085); intranasal MSCs; FUS-assisted intranasal focusing (Ye 2020, PMID 32871204)	Preclinical	No quantitative human pontine exposure data; poor spatial selectivity alone
Implants/depots/devices	Intermittent-CED catheter-port systems; ventricular reservoirs and programmable shunts; skull-implanted FUS arrays (Habashy 2026); osmotic pumps	Access hardware clinical; depots not pontine	No cavity for a wafer; infection/revision burden; volume neutrality
Systemic BBB-penetrant pharmacology	Panobinostat PBTC phase 1 + CNS PK (Monje 2023, PMID 37526549; Zhang 2023, PMID 37827699); imipridones	Standard comparator	Efflux ceiling; whole-body toxicity pays for pontine exposure
Radiation-coupled delivery	CED radioisotopes; BNCT salvage in recurrent DMG (Huang 2023, PMID 36821007); FUS + hypofractionated RT (Tazhibi 2024)	Radiation itself mature; couplings early	Shared oedema/tolerance budget; re-irradiation constraints (Hug 2026, PMID 41488407)

3. What determines success, independent of route

1. **Volume of distribution vs volume infused (Vd/Vi)** and infusion rate — the core CED parameters (Rechberger 2020, PMID 31896090), also implicitly limiting FUS and CSF routes.
2. **Reflux along the catheter track** and trajectory choice through safe entry zones (Mukherjee 2020, PMID 30136133 and the safe-entry-zone atlases).

3. **Residence time and clearance** — the failure mode Power 2023 identified and depots (Bellat 2020) attack.
4. **Efflux status of the payload** — determines whether opening the barrier is sufficient.
5. **Verification:** MR-visible surrogates (Krauze 2005, PMID 16181805; Saito 2005, PMID 16197944), radiolabelled agents (Kommidi 2018, PMID 29456798; ¹²⁴I-omburtamab), and cellular biodistribution imaging (Li 2026, PMID 42068336). Unverified delivery is the reason so many negative DIPG trials are uninterpretable.
6. **Repeatability** in a child with months of life: hardware (ports, reservoirs, implanted arrays) matters as much as pharmacology.

4. What related diseases contribute

Chronic ICV/intrathecal dosing in children is an established, years-long practice in CLN2, MPS II, Sanfilippo, SMA and CLN7 (Hammon 2021, PMID 34076336; Seo 2023, PMID 37922836; Greenberg 2026, PMID 41314141; Kwon 2026, PMID 41360995), and direct parenchymal infusion under MRI guidance was developed for Gaucher disease and AADC deficiency (Lonser 2007, PMID 17065591; Pearson 2021, PMID 34253733; intraputaminally- vs-ICV comparison in Compton 2023, PMID 36472076). What transfers is operational: device longevity, infection and immunogenicity management, and route-comparison methodology. What does not transfer is the therapeutic geometry — these diseases want wide, low-concentration exposure of non-cytotoxic agents, while a pontine tumour wants focal, high-concentration exposure of agents that damage tissue if misplaced. See `notes/10_related_diseases_lessons.md`.

5. Gaps this collection makes visible

- **Verified dosimetry is rare.** Only 5 records classify primarily as imaging/dosimetry/modelling; patient-specific, prospectively validated pontine infusion planning (tract- and deformation-aware, with measured Vd/Vi) does not exist in this literature.
- **No head-to-head route comparisons in DIPG.** The only controlled route comparisons come from gene therapy in other diseases.
- **Residence time is under-engineered.** One clinical strategy (intermittent CED) and a handful of preclinical depots address the field's own stated limiting factor.
- **Combined-route protocols are conceptually obvious and empirically absent:** e.g. CED or FUS for the pontine bulk plus a CSF route for disseminated/leptomeningeal disease, with efflux inhibition for the payload.
- **Posterior-fossa FUS engineering** (skull aberration, implanted arrays, acoustic monitoring in the brainstem) is the highest-leverage technical gap, because it is the only non-invasive repeatable route.
- **Toxicity of local success.** Locoregional CAR-T shows that delivering enough agent to the pons creates its own dose-limiting inflammation; delivery research needs paired oedema-management strategies rather than better delivery alone.

6. Practical reading order

Start with `notes/00_overview.md`, then 01 (CED) and 02 (FUS) for the two routes with clinical momentum, 05 for cellular therapy where the route has already changed practice, 08 for why systemic dosing underperforms, and 09 for how delivery is verified.

Overview: delivering therapeutics to the pons

Why the pons is a delivery problem, not only a drug problem

Diffuse intrinsic pontine glioma (DIPG) / H3K27-altered diffuse midline glioma (DMG) is the paradigmatic "right drug, wrong place" disease. Three properties dominate every delivery decision:

1. **An intact or near-intact barrier.** Unlike glioblastoma, DIPG typically shows little contrast enhancement, and tissue-level work confirms a largely preserved neurovascular unit: El-Khouly 2021 (PMID 37284626) characterised the NVU in DIPG biopsy/autopsy material against age-matched pons; Haumann 2020 (PBTC/SNO abstract PATH-04) compared end-stage DMG pons to healthy controls. Consequently systemic exposure is the exception, not the rule, and drug concentrations achieved in the tumour are usually unmeasured.
2. **Active efflux.** ABCB1/ABCG2 at the pontine BBB restrict exactly the classes of agents that are otherwise active in vitro. Efflux inhibition changes the outcome of delivery experiments: Tsvankin 2020 (PMID 31225627) showed ABC-transporter inhibition plus dexamethasone improved CED dasatinib efficacy in H3.3K27M models, and Power's work (2020/2021, PMID 37128506 for the synthesis) frames pontine delivery as a residence time problem — drug that arrives is cleared before it can act.
3. **Anatomical intolerance.** The pons has no expendable volume: infusion, oedema, or immune-effector expansion translate directly into cranial-nerve and respiratory morbidity. Bander 2020 (PMID 31896089) quantified deformational change after brainstem CED; Hollingworth 2022 (PMID 35933568) catalogued infusion-related side-effects in brainstem CED. Safety in the pons is therefore route- and parameter-specific, not merely drug-specific.

Route taxonomy used in this collection

Route	Mechanism	Maturity in pons
Convection-enhanced delivery (CED)	Pressure-driven bulk flow through interstitium; bypasses BBB entirely	Multiple completed phase 1/2 trials in DIPG
Focused ultrasound + microbubbles (FUS-BBBO)	Transient, targeted, non-invasive BBB opening	Preclinical maturity; first pediatric DMG clinical reports
Intra-arterial $\pm$ osmotic BBB disruption	Regional first-pass concentration; mannitol-opened barrier	Historical brainstem series; modern SIACI experience mostly supratentorial
Intraventricular / intrathecal / intra-CSF	CSF as distribution compartment; ports and reservoirs	Strong for CSF-adjacent and disseminated disease; direct 4th-ventricle infusion modelled in large animals
Locoregional cellular therapy	Cells as both drug and carrier, delivered ICV or intratumourally	Phase 1 in DMG (B7-H3, GD2)
Viral vectors / oncolytic viruses	Intratumoural injection or vector tropism	Phase 1 DIPG (DNX-2401, Ad-TD-nsIL12)

Route	Mechanism	Maturity in pons
Nanocarriers	Improve circulation/uptake, enable co-delivery, or act as depots for CED	Preclinical, plus nanocarrier+CED and nanocarrier+FUS combinations
Intranasal	Olfactory/trigeminal bypass of the BBB	Preclinical (nanoliposomal SN-38, MSCs), enhanced by FUS
Implants, depots, pumps, ports	Sustained local exposure; repeat access without repeat surgery	Extrapolated from GBM wafers/hydrogels; pontine use limited to catheter/port systems
Antibody conjugates and BBB shuttles	Receptor-mediated transcytosis or target-directed payloads	IL13R α 2 ADC/immunotoxin programmes; shuttle work largely non-pontine
Systemic BBB-penetrant pharmacology	Chemistry (Kp,uu, efflux avoidance) rather than device	Underpins ONC201, panobinostat, radiosensitiser trials

Cross-cutting determinants of success

Nearly every paper in this collection can be read as an intervention on one of six variables, and this is the most transferable finding of the review:

- **Where the drug enters** (parenchyma, artery, CSF, nose, blood).
- **How much volume is delivered and how fast** — infusion rate, Vd/Vi ratio, reflux along the catheter track, backflow into CSF spaces.
- **How long it stays** — residence time, efflux, clearance; addressed by depots (Bellat 2020, PMID 32301996), repeat/intermittent infusion (Szycho 2021, PMID 33575829), and efflux blockade.
- **Whether distribution can be seen** — co-infused gadolinium and MR-visible liposomes (Saito 2005, PMID 16197944; Krauze 2005, PMID 16181805) make CED auditable; FUS BBBO uses contrast-enhancement and acoustic feedback.
- **What the anatomy tolerates** — safe entry zones, DTI-defined tracts, and the deformation/oedema budget of the pons.
- **Whether the target is focal or disseminated** — up to half of DIPG cases show leptomeningeal/subventricular spread; a purely focal route cannot address that compartment, which is the main argument for combined CSF + parenchymal strategies.

How records were selected

See `../README.md`. In brief: brainstem/DIPG-specific delivery papers, plus generalisable CNS delivery-technique papers, plus non-oncologic CNS diseases where a direct administration route (catheter, CSF port, FUS, intra-arterial) has clinical experience that transfers to the pons.

Convection-enhanced delivery (CED) to the brainstem

CED is the only pontine delivery route with repeated, completed, prospective clinical experience in DIPG. A cannula placed in the pons infuses at low flow rates so that bulk flow (not diffusion) distributes the agent through the interstitium, making the BBB irrelevant to the delivery step.

Preclinical foundation

- **Feasibility and tolerance in the brainstem.** Sewing 2014 (PMID 25263805) established murine brainstem CED of carmustine with acceptable tolerance and a survival signal. Thomale 2009 built a neurological scoring system for the rat brainstem glioma model precisely because functional morbidity, not just survival, determines whether a pontine infusion is acceptable.
- **Infusate parameters matter as much as the drug.** Rechberger 2020 (PMID 31896090) compared infusate parameters for direct brainstem delivery, and Sandberg 2002 (PMID 12187954) showed hyperosmolar mannitol alters CED distribution in rat brainstem — i.e. the tissue's hydraulic properties are a tunable variable.
- **Agent classes delivered by brainstem CED** in models include EZH2 inhibitors (Sasaki 2020, PMID 32674144), PI3K/MEK combinations (Chang 2019, PMID 32642647), dasatinib with efflux blockade (Tsvankin 2020, PMID 31225627), pan-RAS inhibition (Sbergio 2026, PMID 42371788), and radioimmunoconjugates (Luther 2014, PMID 24526309, theragnostic ¹²⁴I-8H9).
- **Toxicology of the infused agent in brainstem tissue is a separate question from systemic toxicology.** Singleton 2018 (PMID 29856296) characterised distribution, clearance and brainstem toxicity of CED panobinostat in juvenile rat and pig, and Singleton 2017 (PMID 28260886) showed pluronic nano-micellar panobinostat delivered by CED prolonged survival in rat brainstem glioma — the preclinical package behind MTX110 trials.
- **Residence time is the recurring failure mode.** Power 2023 (PMID 37128506) argues explicitly that increasing drug-tumour residence time, not merely getting the drug in, is the translational barrier; Bellat 2020 (PMID 32301996) answered this with a CED-delivered peptide-nanofiber reservoir prolonging local DM1 retention.

Clinical experience in DIPG

Study	Agent	Design	Key point
Anderson 2013 (PMID 23240851)	topotecan	children, brainstem CED	early demonstration of intratumoural infusion in diffuse brainstem tumours
Souweidane 2018 (PMID 29914796)	¹²⁴ I-8H9 (omburtamab)	single-centre phase 1 dose escalation	CED feasible; lesion dosimetry measurable by PET
Heiss 2019 (PMID 30544335)	IL13-Pseudomonas exotoxin	phase 1	targeted toxin delivered by CED in children with DIPG
Szychoť 2021 (PMID 33575829)	carboplatin + sodium valproate	intermittent CED into the pons	

Study	Agent	Design	Key point
			repeat/intermittent infusion via implanted catheters, addressing residence time
Mueller 2023 (PMID 37318058)	MTX110 (aqueous panobinostat)	PNOC015 phase 1, repeated CED	repeated dosing through an implanted catheter; drug chosen for CED compatibility
Saito 2025 (PMID 40150841)	nimustine (ACNU)	multicentre phase 2	largest prospective CED experience in children with DIPG

Safety, imaging and mechanics specific to the pons

- Hollingworth 2022 (PMID 35933568) documented infusion-related side-effects during brainstem CED — transient, dose/volume-related neurological events are expected and must be planned for, not treated as protocol failures.
- Bander 2020 (PMID 31896089) measured deformational change in the pediatric brainstem after CED; infused volume physically displaces tissue.
- Distribution must be imaged, otherwise "delivered" is an assumption. MR-visible surrogates (Saito 2005, PMID 16197944; Krauze 2005, PMID 16181805 — including brainstem targets in non-human primates) and the DIPG-specific CED imaging protocol work from the UCL/GOSH programme (DIPG-18, 2022 abstract) provide the tooling.

What CED does not solve

Focal coverage. Infusion covers a volume of distribution around the catheter, while DIPG infiltrates beyond it and disseminates to leptomeningeal/subventricular compartments in a large minority of cases. CED therefore pairs naturally with CSF-directed or systemic strategies rather than replacing them.

Focused ultrasound-mediated BBB opening (FUS-BBBO)

FUS with intravenous microbubbles transiently and reversibly opens the BBB at a targeted focus. For the pons it is attractive because it is non-invasive, repeatable, and can be aimed at a volume rather than a catheter track — but the posterior fossa imposes a harder acoustic problem (skull incidence, transducer geometry) than the supratentorial targets where FUS clinical practice matured.

Brainstem-specific preclinical evidence

- Alli 2018 (PMID 29753957) demonstrated brainstem BBB disruption with FUS and enhanced doxorubicin delivery — the feasibility anchor for the pons.
- Englander 2021 (PMID 33753753) and Ishida 2021 (PMID 33188825) established safety and drug-delivery enhancement in murine pontine glioma / experimental DIPG models.
- Payload studies: panobinostat (Martinez 2023, PMID 37795179), olaparib as a radiosensitiser ('t Hart 2023, PMID 37019285), liposomal payloads (Payne 2024, bimodal confirmation of liposome delivery after BBBO).
- Combination with radiotherapy: Tazhibi 2024 (PMID 38555449) showed FUS-BBBO is safe and feasible with moderately hypofractionated radiotherapy in brainstem DMG models.
- Sonication strategy is a design variable: Gong 2022 (PMID 35476579) compared sonication patterns and microbubble administration strategies for large-volume delivery — relevant because a pontine target needs volumetric, not point, opening.
- FUS also enhances other routes: Ye 2020 (PMID 32871204) characterised FUS-mediated brainstem delivery of intranasally administered agents (FUSIN), combining a BBB-bypassing route with focal targeting.

Clinical translation

- Wu 2025 (PMID 41223245) reported BBB opening with neuronavigation-guided FUS in pediatric patients with diffuse midline glioma — the first-line clinical evidence that pontine/midline BBBO is achievable in children.
- Consensus and programme-level reviews frame the agenda: Parekh 2023 (PMID 36709715, FUS Foundation DIPG/DMG consensus), Chesney 2024 (PMID 38702518), Martinez 2024 (PMID 38000663), Lu 2025 (PMID 39941880, trials of FUS for brain tumours).
- Device evolution is relevant to the pons: skull-implanted ultrasound arrays (Habashy 2026) remove the skull as an attenuator and enable repeated sessions; transcranial MRgFUS engineering reviews (Kamimura 2025) cover targeting and thermal-prediction constraints.

Secondary uses that matter for pontine trials

FUS is not only a delivery tool. Zhang 2026 (PMID 42281802) used FUS to enrich circulating tumour biomarkers in a DMG mouse model — a non-invasive "liquid biopsy

amplifier" for a tumour that is difficult to sample. Sonodynamic therapy (Hutton 2024, PMID 39217635; PpIX-based approaches, Marcus 2024, PMID 38398131) uses the same hardware for a cytotoxic rather than a delivery endpoint.

Open issues

Dose-response of opening (how much albumin/contrast extravasation equals therapeutic concentration) is rarely measured in the pons; the safety envelope for repeated opening adjacent to respiratory nuclei is not established; and BBBO does nothing about efflux once the drug is inside, which is why FUS+efflux-aware chemistry and FUS+nanocarrier combinations recur in this literature.

Intra-arterial delivery and osmotic BBB disruption

The idea is regional pharmacokinetics: inject into the artery that supplies the target so first-pass concentration is high while systemic exposure stays lower, optionally after opening the BBB with intra-arterial hyperosmolar mannitol. For the pons this means the posterior circulation (basilar/vertebral, superselective pontine branches), which is technically the least forgiving vascular territory in the brain.

Brainstem-specific reports

- Fujiwara 1994 (PMID 8108006) — intra-arterial chemotherapy in four brainstem glioma cases; the earliest brainstem-specific series in this collection.
- Hall 2006 (PMID 16314949) — osmotic BBB disruption chemotherapy for diffuse pontine gliomas: the disease-specific attempt at barrier opening plus regional delivery.
- Riina 2010 (PMID 20377982) — balloon-assisted superselective intra-arterial infusion of bevacizumab for malignant brainstem glioma (technical note): flow arrest to maximise local residence in the posterior circulation.
- Carceller 2014 (PMID 24327128) — superselective intracerebral catheterisation to administer oncolytic virotherapy in DIPG; shows the route can carry biologics and cell-based carriers, not only small molecules.
- McCrea 2021 (PMID 34359048) — intra-arterial bevacizumab and cetuximab with BBB disruption in children with high-grade glioma and DIPG: the most substantial modern pediatric experience.

Supporting adult/supratentorial and mechanistic evidence

- Barrier-opening physiology: Siegal 2000 (PMID 10761648) measured the window of barrier opening after osmotic disruption in humans — a delivery-timing parameter, not a binary state. Roman-Goldstein 1995 (PMID 7793380) paired MR findings with cognitive testing after osmotic disruption plus intra-arterial chemotherapy.
- Modern SIACI programmes: Boockvar 2011 (PMID 20964595, MTD of superselective intra-arterial bevacizumab after osmotic disruption), Chakraborty 2016 (PMID 26945581, cetuximab), Patel 2021 (PMID 34601657, repeated SIACI bevacizumab), Uluc 2022 (PMID 35892048, safety of intra-arterial chemotherapy $\pm$ osmotic disruption), Ferreira 2026 (PMID 41043867 and PMID 40044413), and the systematic reviews by Huang 2020 (PMID 32983974) and Pinkiewicz 2022 (PMID 36212394).
- Pediatric non-brainstem CNS tumours: Jahnke 2008 (PMID 18072268) for embryonal/germ cell tumours, Osztie 2001 (PMID 11337321) for progressive optic-hypothalamic glioma — demonstrating tolerability of the approach in children.
- Emerging payloads: intra-arterial radioligand therapy (Sabuzi 2026, PMID 41594316) and biodegradable intra-arterial devices for focal delivery (Kinra 2026).
- CED interaction: Sandberg 2002 (PMID 12187954) showed hyperosmolar mannitol changes CED distribution in the rat brainstem, so osmotic manipulation is not independent of a concurrent parenchymal infusion strategy.

Assessment for the pons

Intra-arterial delivery has the best anatomical logic for a diffusely infiltrating pontine tumour — it perfuses the whole territory rather than one catheter's volume of distribution — and the worst risk profile: posterior-circulation catheterisation, non-selective opening of normal pontine barrier, and dependence on operator-level neurointerventional expertise. Its current role in DIPG is exploratory, and the literature is dominated by small series, technical notes and adult extrapolation. The mechanistic parameters worth carrying forward are the opening window (Siegal), flow-arrest/balloon assistance (Riina) and repeat-dose feasibility (Patel).

Intraventricular, intrathecal and other intra-CSF routes

The CSF route trades parenchymal penetration for compartment coverage. Drug given into the ventricles or thecal sac reaches CSF-adjacent disease and the leptomeninges at high concentration but penetrates only millimetres into solid parenchyma. For DIPG this makes it complementary to CED/FUS: the pons itself is a parenchymal target, whereas the subventricular and leptomeningeal spread seen in a large fraction of cases is a CSF target.

Direct fourth-ventricle infusion

The most pons-adjacent CSF strategy is infusion into the fourth ventricle, developed in large-animal models by Sandberg's group:

- Sandberg 2008 (PMID 18447671) — chemotherapy directly into the fourth ventricle in a new piglet model.
- Sandberg 2012 (PMID 22546032) — the same approach in a nonhuman primate model.

These studies define tolerability of repeated dosing into a ventricle bounded by the dorsal pons and medulla, and they are the translational bridge for administering agents to the surface of a pontine tumour without traversing it.

Access hardware and its failure modes

- Reservoirs/ports: Sandberg 2000 (PMID 10917346) on Ommaya placement technique and complication avoidance; Zairi 2011 (PMID 21107993) on complications of intraventricular chemotherapy devices; Dupoirion 2023 on intrathecal catheter + subcutaneous port.
- Route comparisons: Zhang 2026 (PMID 42234048) meta-analysed Ommaya reservoir versus lumbar puncture for intrathecal chemotherapy; Jiang 2026 (PMID 42239892) reviewed Ommaya-based intrathecal therapy in leptomeningeal metastasis.
- Administration technique changes distribution: Tuohy 2025 (PMID 41080198) showed that the technique of intra-Ommaya administration alters CSF drug distribution — a reminder that "intraventricular" is not a single, reproducible exposure.
- Programmable VP shunts as access devices: McThenia 2022 (PMID 35403968).
- CSF flow patency must be verified first: Patel 2018 (PMID 29170836) used phase-contrast CSF flow MRI to check pathway patency before intra-CSF therapy — directly relevant to pontine/posterior-fossa tumours that obstruct the fourth ventricle.
- Human CSF tracer distribution in the upper cord and brainstem is now measurable (Melin 2023, PMID 38063195), providing the pharmacokinetic map this route lacked.

Payloads delivered intra-CSF in CNS tumours

Topotecan (intraventricular, 50 pediatric patients, DDEL-05 2022 abstract), CDK4/6 inhibitors given metronomically intrathecally in pediatric brain tumour models (MODL-05 2022 abstract), nanoparticle-formulated panobinostat for intrathecal use (Chaudhuri 2020

abstract), radioimmunotherapy with omburtamab, and — most importantly for DMG — CAR-T cells (see 05_cell_and_viral_vector_delivery.md).

Non-oncologic CSF delivery experience worth importing

The neurogenetics field has industrialised this route and its lessons transfer directly: intrathecal AAV9 gene therapy in CLN7 (Greenberg 2026, PMID 41314141), intrathecal onasemnogene abeparvovec in SMA (Kwon 2026, PMID 41360995), intrathecal idursulfase-IT in MPS II (Muenzer 2026, PMID 41737901), long-term intraventricular enzyme replacement in CLN2 including the practical shift from ventricular access device to chest-sited port (Gopalka 2026, PMID 41897078), and nurse-led intrathecal tofersen clinics in ALS (Jewett 2026, PMID 42438447). Together they show that repeated, protocolised CSF dosing in children is operationally routine — the constraint in DIPG is the drug and the parenchymal barrier, not the access.

Device infection, CSF leak and immunogenicity are the recurring complications in that literature rather than parenchymal toxicity, and glymphatic/perivascular transport work is included where it explains how CSF-borne agents move into brainstem tissue.

Locoregional cellular therapy and viral vectors

For cells and viruses the delivery question is different from small molecules: the agent replicates, migrates and traffics, so the route determines the starting distribution and the systemic toxicity profile rather than the whole exposure.

CAR-T: the route moved intracranial

The clearest recent shift in DMG therapeutics is from intravenous to locoregional (intratumoural / intracerebroventricular, ICV) CAR-T administration via an indwelling ventricular catheter and reservoir.

- **B7-H3, ICV (BrainChild-03, Seattle Children's).** Vitanza 2025 (PMID 39775044) is the completed phase 1 of ICV B7-H3 CAR-T for DIPG; Ronsley 2026 (PMID 42503899) extends the same delivery platform to non-pontine DMG and other recurrent pediatric CNS tumours. Ronsley 2026 (PMID 41798119) describes tumour inflammation-associated neurotoxicity in DIPG patients receiving B7-H3 CAR-T — the pons-specific toxicity that locoregional delivery creates, since local effector expansion happens inside a brainstem with no room.
- **GD2, IV then ICV (Stanford).** Monje 2025 (PMID 39537919) reports intravenous plus intracranial GD2-CAR T cells for H3K27M+ DMG; Jiang 2026 (PMID 40682569) reports a phase 1 GD2-CAR trial with mechanistic analysis in H3K27M+ DIPG. The IV-then-ICV sequence is itself a delivery design: systemic priming, then compartment dosing.
- **Other locoregional programmes.** Gust 2025 (PMID 40070357) reports locoregional EGFR806-CAR T infusion for recurrent/refractory pediatric CNS tumours (BrainChild-01).
- **Making delivered cells go where they are needed.** Vitanza 2025 (CXCR3-A engineering to enhance trafficking of ICV-delivered B7-H3 CAR-T) and Timpanaro 2026 (PMID 41639009, B7-H3 CAR-T + ONC206 in DIPG) treat trafficking and combination as delivery-adjacent problems. Chen 2026 shows anti-CAR immunity driving acquired resistance to GD2-CAR — repeat dosing through a port has an immunological ceiling.
- **Measuring where cells went.** Li 2026 (PMID 42068336) uses dual-modality imaging for longitudinal biodistribution of ICV CAR-T — the cellular analogue of gadolinium co-infusion in CED.
- **Route can be arterial too.** Wang 2024 (PMID 38157532) enhanced CAR-T brain entry using intra-arterial NEO100, linking cell therapy to the intra-arterial literature.

Cells as carriers

- Chastkofsky 2021 (PMID 33272983) — mesenchymal stem cells deliver oncolytic virotherapy to DIPG; the intranasal MSC route was reported earlier by the same group (2017 abstracts) as a non-invasive way to home carriers to the brainstem.
- Chai 2025 (PMID 41062177) — neural stem cell-delivered oncolytic virus given ICV in glioblastoma models, combining a CSF route with a cellular carrier.

- Carceller 2014 (PMID 24327128) — MSC-carried oncolytic adenovirus delivered by superselective intra-arterial catheterisation in DIPG.

Oncolytic viruses and vectors in the pons

- **DNX-2401 (Delta-24-RGD)**. Gállego Pérez-Larraya 2022 (PMID 35767439) — phase 1 intratumoural DNX-2401 into the pons after biopsy, the landmark demonstration that a replicating virus can be injected into a DIPG with acceptable safety; reviewed for brainstem gliomas in Gállego Pérez-Larraya 2023 (PMID 37061388).
- **Ad-TD-nsIL12**. Qian 2025 (PMID 40721414) reports two phase 1 trials in primary and progressive pediatric IDH-wildtype DIPG (the adult HGG counterpart is Ning 2024, PMID 39516192).
- **Combination with immune agonists** (Labiano 2025, CD40 costimulation with Delta-24-RGD in DMG) and with engineered T cells (Vazaios 2025, $\gamma\delta$ -TCR T cells plus oncolytic virus) are the current direction; these change what the injected volume achieves rather than how it is delivered.
- **AAV vectors** matter mainly as a route-comparison resource: whole-brain biodistribution after ICV injection across serotypes (Ye 2026, PMID 42518515), intrathecal AAVrh10 CNS gene transfer (Hordeaux 2015, PMID 25588740), and the intraputamina-versus-ICV comparison for AAV2-hAADC (Compton 2023, PMID 36472076) — direct evidence that parenchymal infusion and CSF injection produce different distributions of the same vector. A reported neuroepithelial tumour with AAV integration after intracisternal magna delivery (Ahrens-Nicklas 2026) is the safety counterweight.

Practical implications

1. An implanted ventricular catheter/reservoir turns a one-shot operation into a dosing platform; nearly all pontine cell-therapy progress depends on that hardware.
2. Locoregional dosing shifts toxicity from systemic CRS to local inflammation, which in the pons is the dose-limiting phenomenon.
3. Delivery success for cells is measured by trafficking and persistence, so imaging and CSF biomarker sampling belong in the delivery protocol.

Nanocarriers and the intranasal route

Both of these are best understood as modifiers of the routes in the other notes rather than standalone pontine delivery solutions: a nanocarrier changes residence time, solubility and uptake of whatever route carries it, and the intranasal route provides a non-invasive entry point that is then focused by ultrasound or by cellular homing.

Nanocarriers in DIPG/DMG

- **Formulation solves solubility, not location.** MTX110 (water-soluble panobinostat) and nanoliposomal SN-38 exist because the free drugs cannot be delivered at the required local concentration; both are used with a route (CED for MTX110, intranasal for nanoliposomal SN-38, Sasaki 2023, PMID 36599085 — see below). Nano-micellar formulation can also be the enabler of a route: Singleton 2017 (PMID 28260886) prolonged survival with CED of panobinostat-loaded pluronic nano-micelles in rat brainstem glioma.
- **Carrier as depot for CED.** Bellat 2020 (PMID 32301996) combined CED with a peptide-nanofiber reservoir to prolong local DM1 retention — the clearest example of nanotechnology addressing the residence-time failure mode.
- **Carrier as payload for FUS.** Payne 2024 (PMID 39624326) confirmed liposome delivery to brain after FUS-BBBO with two independent modalities, defining the time window in which large carriers can cross.
- **Targeted and gene-silencing carriers in DIPG models.** Functionalised macrophage exosomes co-delivering panobinostat and PPM1D-siRNA (Shan 2022, PMID 35585670), albumin-functionalised gold nanoparticles for doxorubicin (Ung 2020 abstract), liposomal nanocarriers for targeting and optical imaging of brainstem tumours (Kambhampati 2014 abstract), spherical nucleic acids against IDO1 (Lenzen 2017 abstract), and myeloid-reprogramming nanoparticle resiquimod (McSwain 2026).
- **Reviews specific to the disease:** Bredlau 2017 (PMID 26903150, nanotechnology applications for DIPG), Presswala 2024 (PMID 39091904), Mohamed 2026 (PMID 42307690, DMG nanoparticle prospects), Alaseem 2025 (PMID 40828479, BBB-crossing strategies in pediatric CNS tumours).
- **Radiation interaction.** Metallic nanoparticles as radiosensitisers with distance-dependent Auger effects in DMG (Al-Husseini 2026, PMID 42394610) and local delivery of radiosensitising nanoparticles to pediatric brainstem gliomas (2016 abstract) tie the nanocarrier literature to the radiotherapy backbone that every DIPG patient receives.
- **Extracellular vesicles cut both ways:** they are candidate carriers, but in DMG they also mediate radioresistance signalling (Oza 2025, PMID 41167985; Morcos 2026, PMID 42352467), so an EV-based delivery platform is not biologically neutral.

Caveat: the majority of nanocarrier records in this collection are preclinical or platform reviews. No nanocarrier has a completed pontine-specific clinical delivery trial; where nanomedicine has reached DIPG patients it has done so through CED (MTX110) or as a formulation of a conventional agent.

Intranasal / nose-to-brain

- **DIPG-specific preclinical work.** Sasaki 2023 (PMID 36599085) — intranasal nanoliposomal SN-38 for DMG (preceded by He 2017/2018 abstracts on brainstem glioma). Chastkofsky 2017 (abstracts, and 2021 full paper for the MSC/virus combination) — intranasal MSCs home to DIPG in mouse models.
- **Focusing the route.** Ye 2020 (PMID 32871204) and related FUSIN work combine intranasal administration with focused ultrasound to concentrate agents in the brainstem — the intranasal route supplies drug to the CNS broadly, and FUS supplies the spatial selectivity the nose cannot.
- **Transferable formulation science** comes largely from neurodegeneration: chitosan and lipid nanocarriers, nasal deposition targeting the olfactory region, and hydrogel vehicles (see `10_related_diseases_lessons.md`). Clinical evidence for nose-to-brain delivery in humans remains limited; the DIPG-relevant records are all animal studies.

Judgement

For the pons, nanocarriers are most credible where they extend local residence after a direct infusion, or where they are the only way to formulate an otherwise undeliverable agent. Intranasal delivery is attractive for repeated, non-invasive dosing in children but currently lacks quantitative pontine exposure data in humans; the FUS-assisted variant is the most plausible clinical path.

Implants, depots, pumps and access devices

Two distinct ideas share this category:

1. **Depots** that release drug locally over time (wafers, hydrogels, microspheres, nanofiber reservoirs, osmotic pumps).
2. **Access hardware** that makes repeated delivery possible without repeated surgery (indwelling brainstem catheters with subcutaneous ports, ventricular reservoirs, programmable shunts, implanted ultrasound arrays, implanted intrathecal ports).

For the pons, only the second idea is currently clinical — there is no resection cavity to line with a wafer — but the depot literature is the source of the residence-time solutions the pontine field needs.

Access hardware relevant to the pons

- Implanted catheter + port systems enabling intermittent CED into the pons (Szycho 2021, PMID 33575829; the Bristol/UCL intermittent-CED phase 1 abstracts, DDEL-07 2020 and SURG-07 2024) — the pontine analogue of an Ommaya reservoir.
- Ventricular reservoirs and programmable VP shunts as dosing platforms for ICV therapy (McThenia 2022, PMID 35403968; Zairi 2011, PMID 21107993; Sandberg 2000, PMID 10917346).
- Implantable intrathecal catheter-ports designed for chronic dosing, now being evaluated prospectively in SMA (ThecaFlex DRx, Mesina-Estarrón 2026, PMID 41830093) and adopted in CLN2 (chest-sited port instead of a ventricular access device, Gopalka 2026, PMID 41897078).
- Skull-implanted ultrasound arrays for repeated BBB opening (Habashy 2026, PMID 42281806), which convert FUS from a session-based procedure into an implanted delivery device.
- Implantable intracerebral osmotic pumps for continuous convection-enhanced delivery (Berlet 2025, PMID 41132728), addressing the duration limit of a single infusion session.

Depots and local-release matrices (mostly glioblastoma-derived)

- Carmustine (Gliadel) wafers remain the only FDA-approved implanted intracranial chemotherapy; the collection includes both effectiveness/complication series (Gkasdaris 2024, PMID 39456559; Houedjissin 2026, PMID 42187555; Roux 2025, PMID 40971079; Suryadevara 2025, PMID 41030714) and the complication literature that constrains any pontine analogue — CSF leak, eosinophilic meningitis, oedema and cyst formation (Kawaguchi 2024, PMID 38841004; Kuramitsu 2015, PMID 28663961). Kartal 2026 (PMID 41899511) reviews the limitations and next-generation directions.
- Injectable/in-situ hydrogels carrying nanoparticles, cells or immune agonists (Morshedi 2026, PMID 41116065; Maue 2026, PMID 41957222, thermoresponsive iNSC scaffold; Shang 2026, PMID 41499494, STING prodrug gel; Yadav 2025, PMID 41184309, conductive gels for electro-immunotherapy) show how a depot can co-deliver several agents at once, but all assume a cavity or an injectable volume that the pons cannot spare.

- Microneedle and other minimally invasive device concepts appear as reviews only; no brainstem application yet.
- CED-compatible depots are the crossover point: peptide nanofiber reservoirs (Bellat 2020, PMID 32301996), nano-micellar panobinostat (Singleton 2017, PMID 28260886), and self-assembling polypeptide-drug conjugates (DDEL-02 2024 abstract for an IL13R α 2-targeted "self-depot" in GBM and DIPG).

Design constraints a pontine device must satisfy

- Volume neutrality: displacement of even ~1 mL matters (Bander 2020, PMID 31896089).
- Catheter track must respect safe entry zones and long tracts (see 09_surgical_access_imaging_and_modeling.md).
- Infection and CSF-leak risk over months of repeated access — the dominant complication class in both the Ommaya and Gliadel literature.
- Compatibility with MRI-based distribution monitoring and with radiotherapy planning, since almost all patients are irradiated.

Pontine BBB biology, efflux, and systemic BBB-penetrant pharmacology

Every route decision rests on an empirical question: how much of the barrier is actually intact in a given DIPG, and what does it do to the specific molecule?

What is known about the barrier in DIPG/DMG

- **Structure.** El-Khouly 2021 (PMID 37284626) examined the neurovascular unit in DIPG biopsy and autopsy tissue against age-matched pons; Haumann 2020 (PATH-04 abstract) compared end-stage DMG pons to healthy controls, framing the "intact BBB" claim in tissue terms rather than by imaging inference alone.
- **Regulation.** Wei 2021 (HGG-22 abstract) reported differences in BBB function and increased Angpt1 in H3K27M DIPG models, i.e. the tumour may actively maintain barrier integrity — a delivery-relevant mechanism, not an incidental property.
- **Heterogeneity.** Contrast enhancement in DMG is variable and changes after radiotherapy (Rameh 2025, PMID 40280708), so a single patient's barrier state is a moving target across the treatment course.

Efflux transporters set the ceiling for systemic dosing

- ABCG2/ABCB1 promote brainstem gliomagenesis and blunt dasatinib efficacy in DIPG models (2014 abstract; Tsvankin 2020, PMID 31225627 for the therapeutic corollary).
- Power's CED/efflux work (HGG-13 2021 and DDEL-16 2020 abstracts; Power 2023, PMID 37128506) shows brain distribution and clearance of alisertib depend jointly on P-gp/BCRP and on the delivery method — the two variables are not separable.
- Practical consequence: an agent's in vitro potency in H3K27M cells predicts little; the relevant quantities are unbound brain concentration, $K_{p,uu}$, and residence time.

Pharmacokinetic evidence generated for pontine trials

- Panobinostat: PBTC phase 1 in children with DIPG (Monje 2023, PMID 37526549), CNS distribution modelling to guide pediatric dosing (Zhang 2023, PMID 37827699), non-human primate blood-versus-CSF levels (EPCT-09 2021 abstract), and a ^{18}F -radiolabelled analogue enabling PET-guided delivery assessment (Kommidi 2018, PMID 29456798). This is the most complete example in the field of pairing a drug with delivery measurements.
- ONC201/ONC206 (imipridones) are used precisely because they are BBB-penetrant; the collection includes resistance mechanisms (Jackson 2020 abstract, AKT-driven resistance) and combination work with locoregional CAR-T (Timpanaro 2026, PMID 41639009).
- Radiosensitiser and metabolic programmes assume systemic dosing plus radiotherapy (Shen 2025, PMID 39575457, on mitochondrial bioenergetics/hypoxia; Liu 2024, PMID 38644565, anlotinib with RT/TMZ in H3K27M-DMG).

Interpretation

Systemic pharmacology and physical delivery are complements, not competitors. The literature supports a simple decision rule: if a molecule cannot be engineered to a usable unbound brain concentration at tolerable systemic exposure, it needs a route (CED, FUS, CSF); if it can, the route becomes a way to raise the local ceiling or to shorten the exposure needed. In either case, unmeasured delivery is the main reason DIPG trials cannot distinguish drug failure from delivery failure.

Surgical access, imaging guidance, dosimetry and modelling

Delivery to the pons is executed by a trajectory. This note collects the literature that determines whether a catheter, needle or acoustic focus can be placed where the drug is needed, and whether the resulting distribution can be measured.

Biopsy as the enabling procedure

Molecular diagnosis (H3K27M/H3K27-altered status) made brainstem biopsy routine, and the same trajectory logic now supports therapeutic infusion.

- Role: Sheikh 2024 (PMID 39763605) on brainstem biopsy and targeted therapy in DMG/DIPG; Samadani 2003 (PMID 14742957) for the broader diagnostic indication.
- Safety and yield: Dalmage 2023 (PMID 37724839) on survival and neurological outcomes after DIPG stereotactic biopsy; Dawes 2019 (PMID 30523502) prospective robot-assisted brainstem biopsy in children; Junior 2026 on microsurgical biopsy through safe entry zones; transfrontal transventricular and extraventricular trajectory descriptions (Pereira 2008, PMID 18686061; Amundson 2005, PMID 15796398).
- Biopsy also has a differential-diagnosis function that protects against delivering cytotoxics into a non-neoplastic pontine mass (inflammatory and demyelinating lesions can mimic DIPG radiologically).

Trajectories, safe entry zones and neuronavigation

- Safe entry zones are the best-characterised part of pontine access: high-resolution diffusion-tensor mapping (Mukherjee 2020, PMID 30136133), microsurgical and morphometric atlases (Cavalcanti 2016, PMID 26452114; Yagmurlu 2014, PMID 24983443; ultrahigh-resolution 7T anatomy in Hanalioglu 2024, PMID 39664682; Guberinic 2022, PMID 35395628; Serrato-Avila 2022, PMID 34715371), virtual/3-D exploration (Tayebi Meybodi 2020, PMID 32474103) and outcome comparisons between zones (Catapano 2023, PMID 36681989).
- Approach selection for lateral/central pons — trans-middle cerebellar peduncle (Graffeo 2024, PMID 37976511), lateral pontine zone (Cavalcanti 2020, PMID 32442280), peritrigeminal presigmoid retrolabyrinthine (Hoz 2023, PMID 37578224), endoscopic endonasal transclival (Weiss 2019, PMID 29750275) — comes largely from the cavernous malformation literature but defines the corridors available to a catheter.
- Neuronavigation in brainstem glioma surgery (Zhang 2023, PMID 37205194) shows the practical toolchain; FUS is now neuronavigation-guided as well (Wu 2025, PMID 41223245).

Seeing the delivered agent

- **CED:** MR-visible surrogates — gadoteridol-loaded liposomes visualised in real time in the monkey brain including brainstem (Krauze 2005, PMID 16181805; Saito 2005, PMID

16197944) — plus DIPG-specific imaging protocols for assessing infusate distribution and tumour coverage (DIPG-18 2022 abstract, Szychot/GOSH programme).

- **Radiolabelled agents:** ^{124}I -8H9 theragnostic CED (Luther 2014, PMID 24526309) and ^{18}F -panobinostat PET (Kommidi 2018, PMID 29456798) allow dosimetry rather than assumption.
- **Cells:** dual-modality longitudinal biodistribution imaging of ICV CAR-T (Li 2026, PMID 42068336).
- **CSF routes:** phase-contrast CSF flow MRI to confirm pathway patency before intra-CSF therapy (Patel 2018, PMID 29170836) and human CSF tracer distribution in brainstem and upper cord (Melin 2023, PMID 38063195).

Modelling and dosimetry

- Infusate-parameter studies (Rechberger 2020, PMID 31896090) and mannitol/CED interaction (Sandberg 2002, PMID 12187954) are effectively experimental versions of the transport model.
- Deformation measurement after pontine CED (Bander 2020, PMID 31896089) provides the mechanical constraint that any patient-specific simulation must reproduce.
- Radiotherapy-side dosimetry appears where it interacts with delivery: FUS plus hypofractionated RT (Tazhibi 2024, PMID 38555449), brachytherapy-style permanent implants (GammaTile dosimetry, Zhang 2024), and radiation-volume implications of extrapontine progression (DIPG-52 2023 abstract).

Gap

Patient-specific, prospectively validated pontine infusion planning (tract-aware, deformation-aware, with measured V_d/V_i and reflux) is not established in this literature: modelling records are the smallest category in the index, and most distribution knowledge is empirical and centre-specific.

Lessons from related diseases

These records were kept only where a direct delivery route (CED/parenchymal infusion, CSF, focused ultrasound, intra-arterial, implanted device) was used in humans or large animals for a non-DIPG disease, because the operational knowledge transfers even when the biology does not.

Chronic CSF dosing in children is an established practice

- **CLN2 (cerliponase alfa)** is the strongest precedent: ICV enzyme replacement in children via an implanted reservoir, with dose selection translated from animal models (Hammon 2021, PMID 34076336), immunogenicity monitoring over repeated dosing (Cherukuri 2018, PMID 30205177), long-term real-world practice (de los Reyes 2020, PMID 32684372; Soangra 2025, PMID 39762319) and hardware evolution from ventricular access device to chest-sited port (Gopalka 2026, PMID 41897078).
- **MPS II / Sanfilippo** replicate the pattern with idursulfase-beta and traletsinidase alfa (Sohn 2018, PMID 29978271; Seo 2021, PMID 33768130; Seo 2023, PMID 37922836; Pinkstaff 2023, PMID 36923444; Ueda 2026, PMID 42496452), including same-day combined intravitreal + ICV dosing (Rogers 2025, PMID 40578219).
- **SMA and CLN7 gene therapy**: intrathecal onasemnogene (Kwon 2026, PMID 41360995) and first-in-human high-dose intrathecal AAV9 for CLN7 (Greenberg 2026, PMID 41314141) show intrathecal vector dosing at clinical scale, with implantable intrathecal ports now in trials.

Transferable lessons: reservoirs/ports can be maintained for years in children; infection and device revision are the main recurring complications; immunogenicity and anti-vector immunity limit repeat dosing; and CSF concentration is not a reliable proxy for parenchymal exposure.

Direct parenchymal infusion outside oncology

- **Neuronopathic Gaucher disease**: Lonser 2005 (PMID 15786474) and Lonser 2007 (PMID 17065591) — convection perfusion and then image-guided direct convective delivery of glucocerebrosidase, the same technique/imaging package as tumour CED but for an enzyme; Meng 2022 (PMID 36089809) later delivered recombinant glucocerebrosidase to the putamen with MR-guided focused ultrasound, i.e. the same payload by two different routes.
- **AADC deficiency and Parkinson disease**: MR-guided intraputaminal AAV2-AADC (Pearson 2021, PMID 34253733), and the AAV2-hAADC comparison of intraputaminal versus ICV delivery (Compton 2023, PMID 36472076) — a controlled route comparison rarely available in oncology.
- These programmes established intraoperative MRI infusion monitoring, real-time distribution feedback and cannula designs that pontine CED subsequently reused.

Focused ultrasound matured in adult neurology first

FUS BBB opening in Parkinson and Alzheimer disease supplied the safety framework, implanted-array hardware and acoustic monitoring standards later applied to DMG; the posterior fossa remains the harder target because of skull geometry and the absence of a tolerance margin for oedema.

What does not transfer

- Non-oncologic targets are usually diffuse (whole cortex, whole cord), so wide, low concentration CSF distribution is acceptable — the opposite of the focal, high-concentration requirement in a pontine tumour.
- Enzyme and ASO products are non-cytotoxic, so extravasation or misdistribution is far less dangerous than with chemotherapy or a replicating virus.
- Treatment horizons are years, allowing device strategies that a 9-12 month median survival cannot justify.

Radiation: the one modality that always reaches the pons

Radiotherapy is the only treatment that reliably deposits its "dose" in a DIPG, so it appears in this collection as the comparator and partner for every other route.

Baseline and re-treatment

- Conventional focal RT remains standard; hypofractionated schedules are non-inferior in meta-analysis (Viani 2024, PMID 39144263), which matters for delivery because it changes how often a child is available for procedures.
- Re-irradiation is widely practised at progression (Mankuzhy 2024, PMID 38430472; Nuryadi 2025, PMID 41148593 systematic review/meta-analysis; the Spanish and other multi-centre abstracts), and proton re-irradiation near the brainstem defines the tolerance envelope any combined-delivery protocol must live inside (Hug 2026, PMID 41488407).
- Craniospinal irradiation appears as rescue for disseminated disease (DIPG-04 2024 abstract) — the radiation analogue of choosing a CSF route over a focal one.

Radiation used to enable or amplify drug delivery

- **Sequencing with BBB opening.** Tazhibi 2024 (PMID 38555449) shows FUS-BBBO is safe and feasible with moderately hypofractionated RT in brainstem DMG — the key demonstration that a physical delivery procedure can be interleaved with the radiation backbone.
- **Radiosensitiser delivery by CED.** Yang 2011 (PMID 20577779) combined CED carboplatin with radiotherapy in an experimental brainstem/glioma model, and DIPG-41 (2024 abstract) reports CED-mediated radiosensitisation for DMG; MTX110 and carboplatin/valproate CED programmes are also designed around the post-RT window.
- **Radiation as the payload.** CED of radio-immunotheranostic ^{124}I -omburtamab (Souweidane 2018, PMID 29914796; full dose-escalation report Souweidane 2025, PMID 39969230, with erratum PMID 40489365) delivers the isotope directly, making dosimetry a delivery measurement. Rhenium-186 liposomes function as convection-delivered brachytherapy (Phillips 2012, PMID 22427110), and permanent implants have measurable dosimetry (GammaTile, Zhang 2024, PMID 38902561).
- **Boron neutron capture therapy** is a delivery problem disguised as radiotherapy: the boron carrier must accumulate in tumour. Salvage BNCT has been given to children with recurrent DMG (Huang 2023, PMID 36821007) and compassionately for brainstem tumours (Chen 2022, PMID 35455057; Liao 2025, PMID 40103665 for clival chordoma with brainstem involvement), and carrier engineering continues (Chen 2026, PMID 41738936, ^{124}I -labelled exosome BNCT platform; Miyata 2011, PMID 21273928, transferrin-targeted boron/iodine liposomes).

Interactions to respect

- Radiation changes the barrier and the tumour vasculature, so contrast enhancement and presumed BBB state after RT are not the same as at diagnosis (Rameh 2025, PMID 40280708).
- Oedema tolerance is the shared limiting resource: FUS-BBBO, CED infusate volume and re-irradiation all draw on it, so combined protocols must budget it explicitly.
- Devices and implants must be radiotherapy-planning compatible; catheter and reservoir hardware affects dose calculation and cannot be ignored in the plan.

Appendix: full reference list

All 1066 curated records, grouped by primary modality and ordered by year (descending).
Records without a PMID are conference abstracts.

convection enhanced delivery (225)

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